Module-2

① Explain feature selection Compare filter and wrapper methods

feature selection :- It is the process of selecting the most relevant & useful features is called feature selection.

In machine learning a data set may contain a large no. of feature not all features are useful for predicting the target variable

Ex:- Suppose we want to predict whether a student will pass an exam. The dataset may

Contain :-

Attendance previous marks
Internal marks student name favourite colour.
study hours Roll number

objectives of feature selection :-

1. Reduce the no. of features in the dataset
2. Remove irrelevant & redundant features.
3. Improve model performance
4. Reduce overfitting

Types of feature selection methods :

1. filter methods
2. wrapper methods
3. Embedded methods

Advantages of feature selection :-

- Reduces dimensionality
- Improves model efficiency
- Reduces overfitting
- Makes models easier to interpret

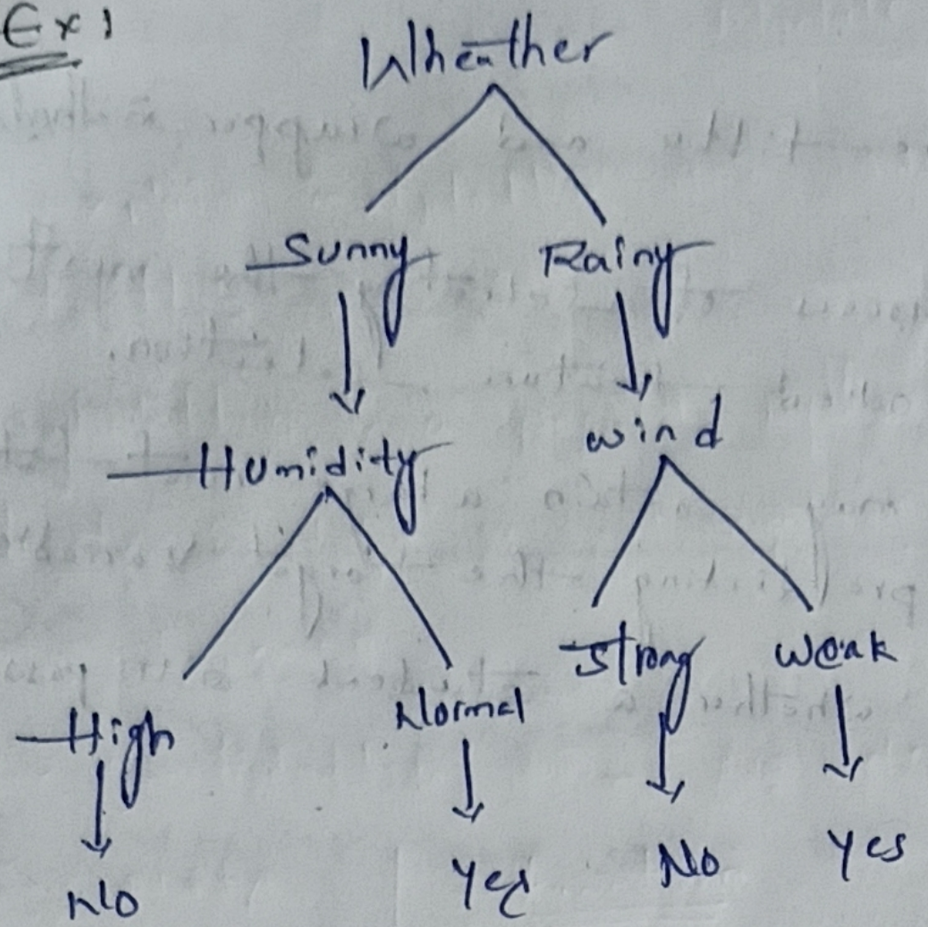
② Describe the decision tree algorithm & its advantages

Decision Tree :-

A decision tree is a supervised machine learning algorithm used for both classification & regression problem the tree consists of

Root Node
Branches
Leaf Node

Ex:



Advantages :-

- Easy to understand
- Easy to interpet
- Handles classification & Regression.

Disadvantages :-

- They can easily overfit training data
- Very deep trees can become Complex.
- large trees may be difficult to implement.

③ Explain principal Component analysis (PCA) & its applications!

PCA :-

It is a dimensionally reduction technique used in machine learning and statistics to radace the no. of features in a dataset while presentation of many features may contain similar or redandant information PCA transforms the orgiand features into a new set of variables

Why PCA Needed :-

→ Increase Compulation Cost
→ Make visualization difficult
→ Increase the possibility of overfitting
→ Make the dataset more complex.

Advantages

1. Reduce dimensionality
2. Reduce computational Complexity

3. Removes Redundancy
4. Helps redudancy
5. Can reduce noise

Applications:-
1. Data visualization
2. Image processing
3. Machine Learning
4. face recognization

Limitations:-
1. It is sensitive of feature scaling
2. PCA can make the orginal features harder to interapt
3. PCA is mainly designed to capture linear relationship